UNO Agent Arena

Machine Learning 2 - Group 1

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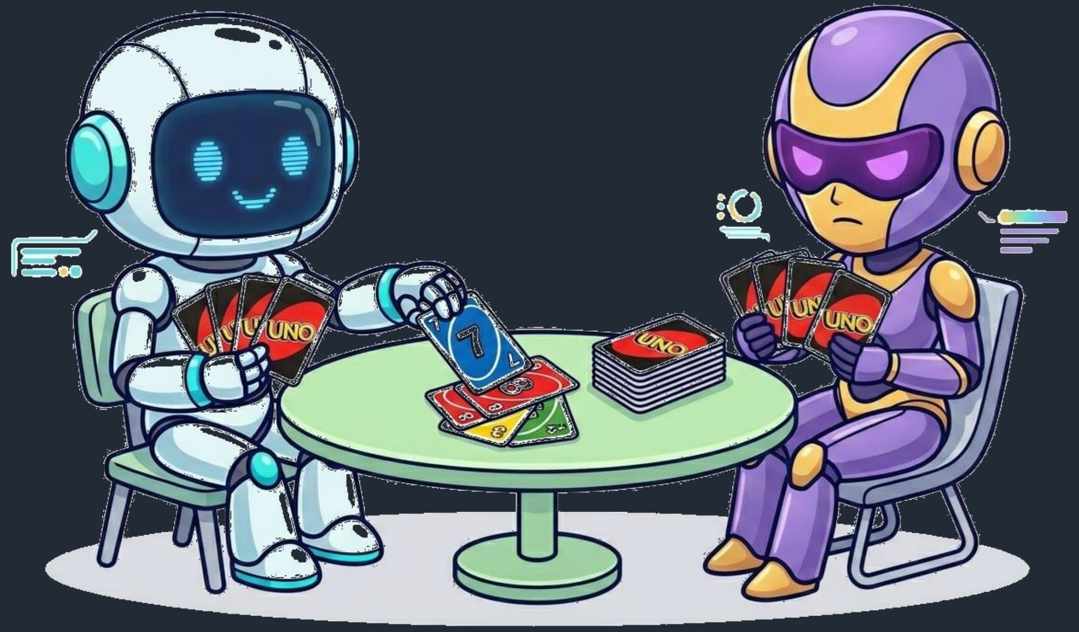
Objectives

- Analyze performance of various agents (algorithms) playing UNO in a 2-player match (one versus one)

- *Is there an unbeatable agent?*

- Can *you* beat our UNO agents?



Contents

Uno rules & Game set up

About The UNO® Card Game

- Originally developed in 1971 by Merle Robbins
- Acquired by Mattel on January 23, 1992
- Available in 80+ countries
- 150+ million copies sold

A standard deck consists of 108 cards

- Four colors: (Red ●, Green ▼, Blue ■, Yellow ✦) symbols for accessibility
- Number Cards: Numbers 0-9, where 0 appears once, 1-9 twice per color
- Action Cards: Draw Two (+2), Reverse (↗↙), Skip (ø), two copies each per color
- Wild Cards: Wild and Wild Draw Four, four copies each
- (112 with added cards like 3x Wild Customizable Cards and 1x Wild Shuffle Hands)

How UNO® Is Played

Setup:

1. Choose a dealer and shuffle cards.
   (No designated dealer in our program)
2. Deal 7 cards to each player.
3. Place the remaining cards **FACEDOWN** in the center of the table. This is the **DRAW PILE**.
4. Flip over the top card of the **DRAW PILE** and place it **FACEUP** to form the **DISCARD PILE**. If this is an Action Card, ignore it and flip over the next card.
5. The player to the left of the dealer goes first and play proceeds clockwise.
   (Player 1 is the starting player in our program)

UNO® Action And Wild Cards

Draw Two (+2):
When played, the next player must draw 2 cards and lose their turn

Reverse (↗↙):
When played, the direction of the play is reversed.
If play was moving clockwise, it now moves counterclockwise and vice versa.
In a two-player match the Reverse works like Skip.

Skip (ø):
When played, the next player loses their turn.

Wild:
This card matches anything so you can play it no matter what card is on the Discard Pile. When you play a Wild card, you also get to choose the color that continues play.

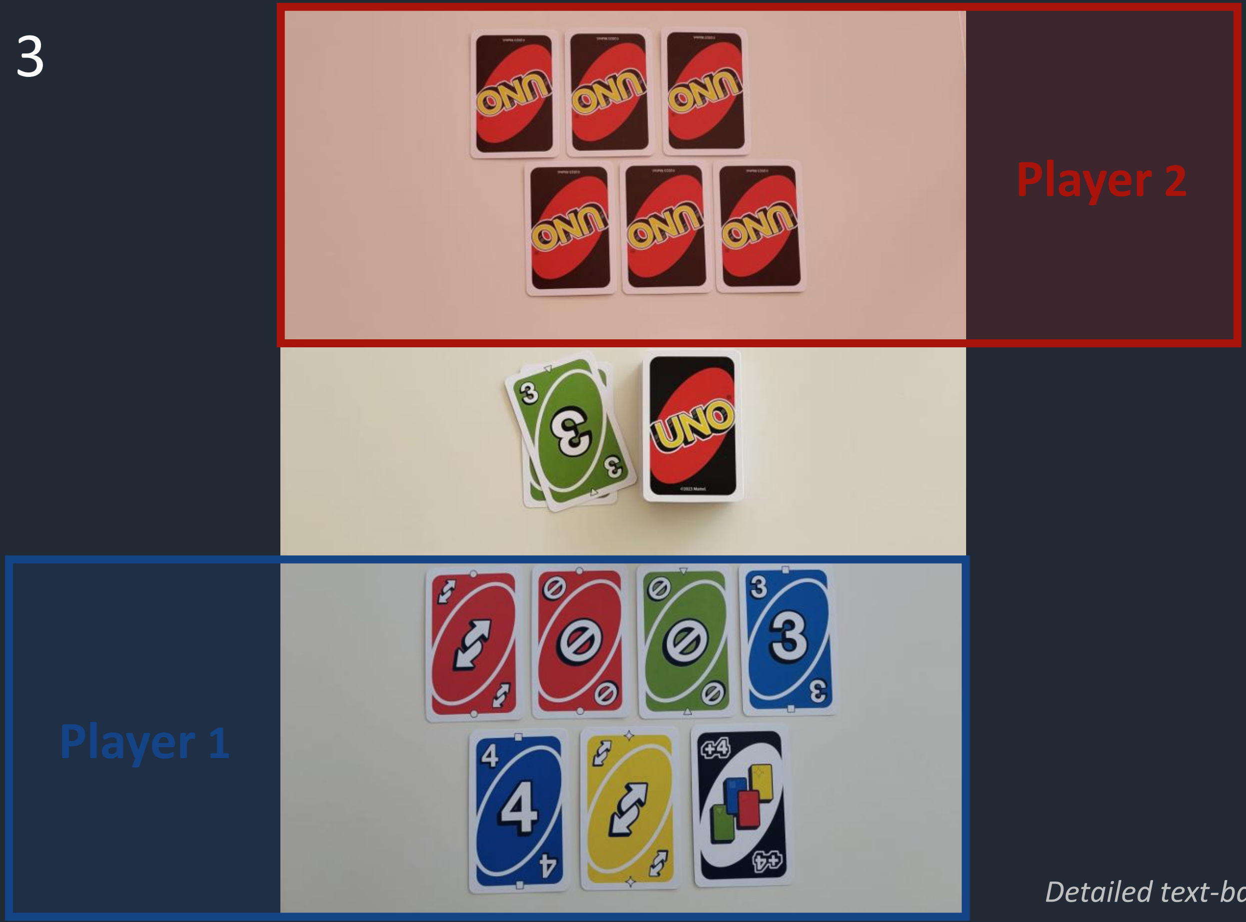
Wild Draw Four (+4):
This card matches anything so you can play it no matter what card is on the Discard Pile. When played, the next player must draw 4 cards and lose their turn.
This card is also a Wild card, so you get to choose the color that continues play.
NOTE: No challenge mechanic! (explained in appendix)

UNO® Gameplay Illustrations

Detailed text-based gameplay description in appendix

UNO® Gameplay Illustrations

Player 2 plays Green 3



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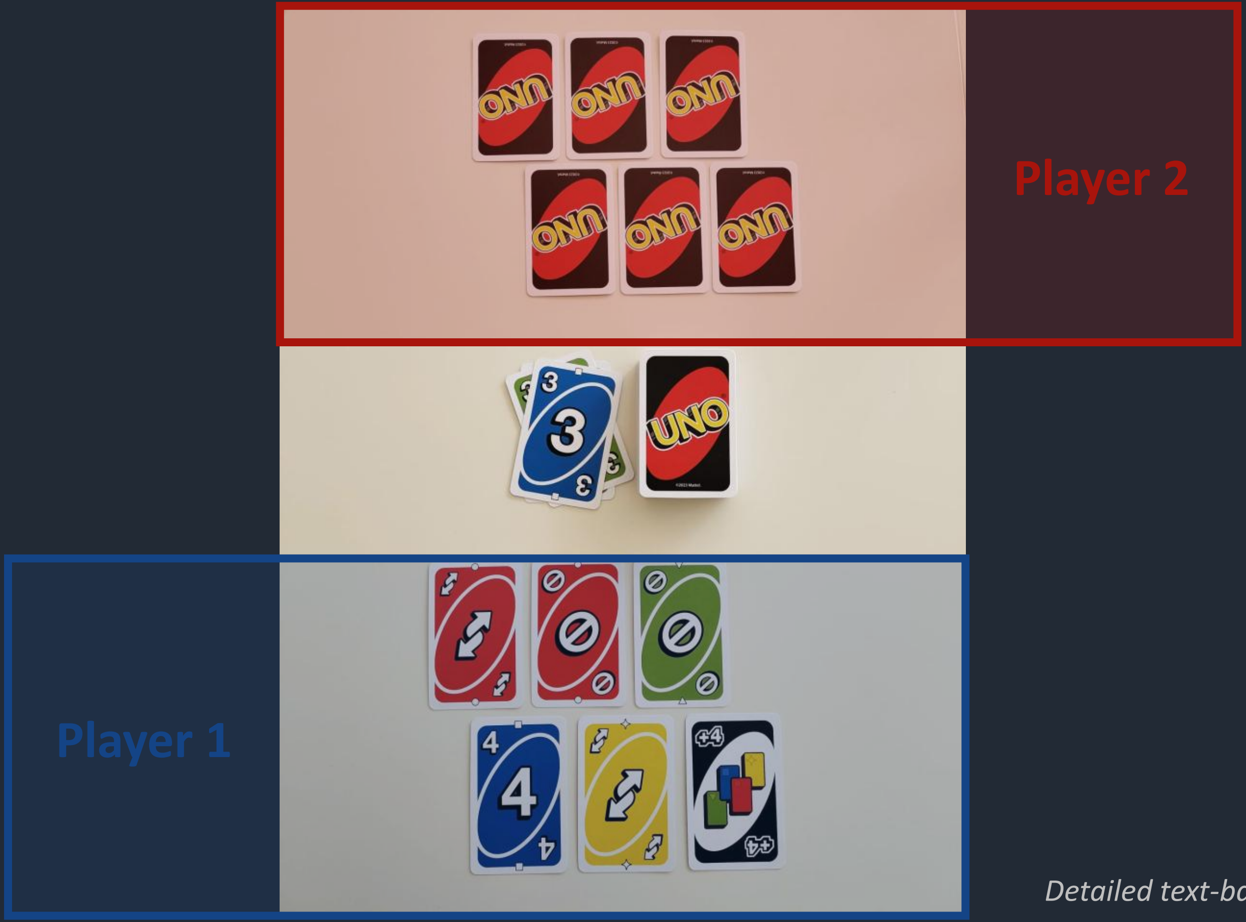
UNO® Gameplay Illustrations

Player 1 plays Blue 3



Detailed text-based gameplay description in appendix

UNO® Gameplay Illustrations

Player 2 plays Blue 7

Detailed text-based gameplay description in appendix

UNO® Gameplay Illustrations

Player 1 plays Blue 4

Detailed text-based gameplay description in appendix

UNO® Gameplay Illustrations

Player 2 plays Yellow 4

Detailed text-based gameplay description in appendix

UNO® Gameplay Illustrations

Player 1 plays Yellow Reverse

Detailed text-based gameplay description in appendix

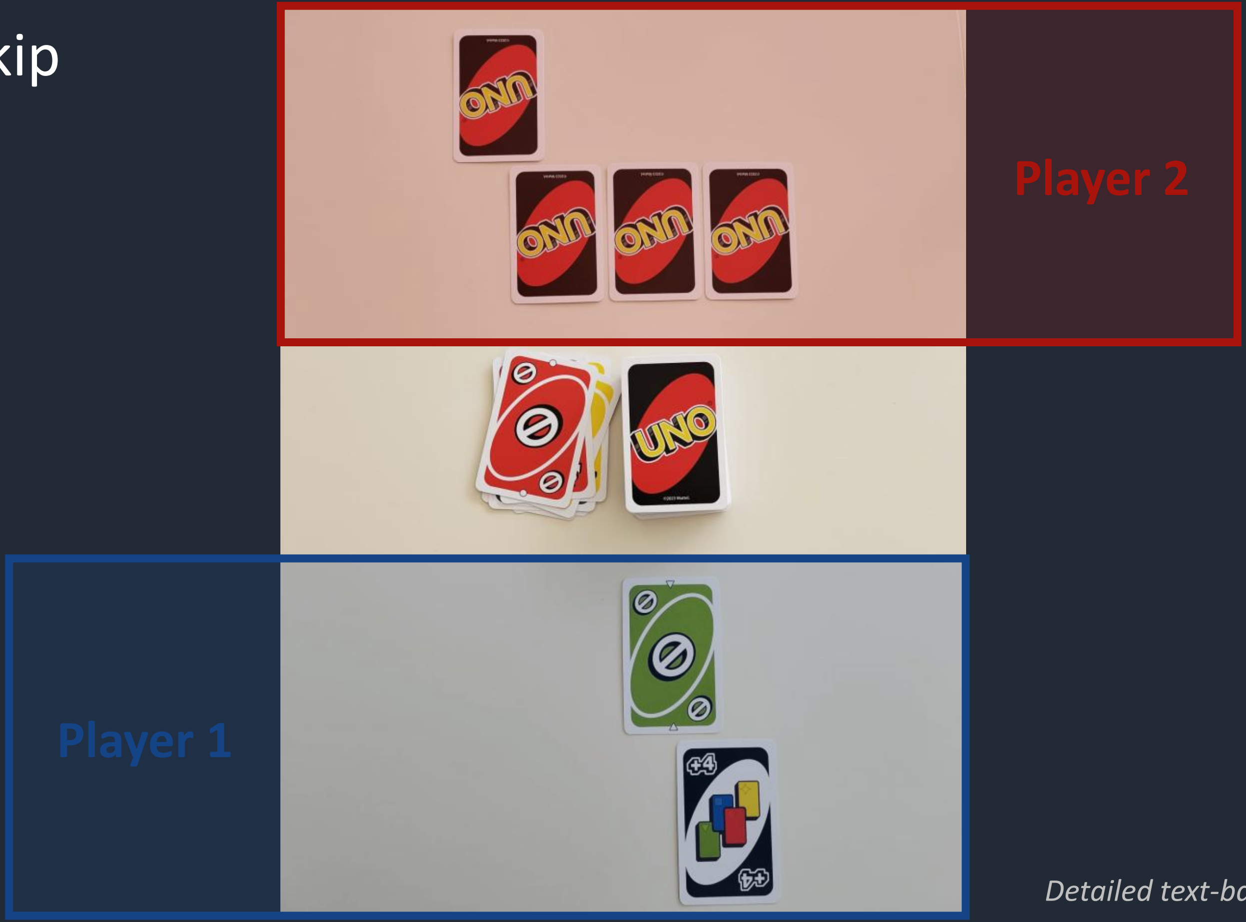
UNO® Gameplay Illustrations

Player 1 plays Red Reverse

Detailed text-based gameplay description in appendix

UNO® Gameplay Illustrations

Player 1 plays Red Skip



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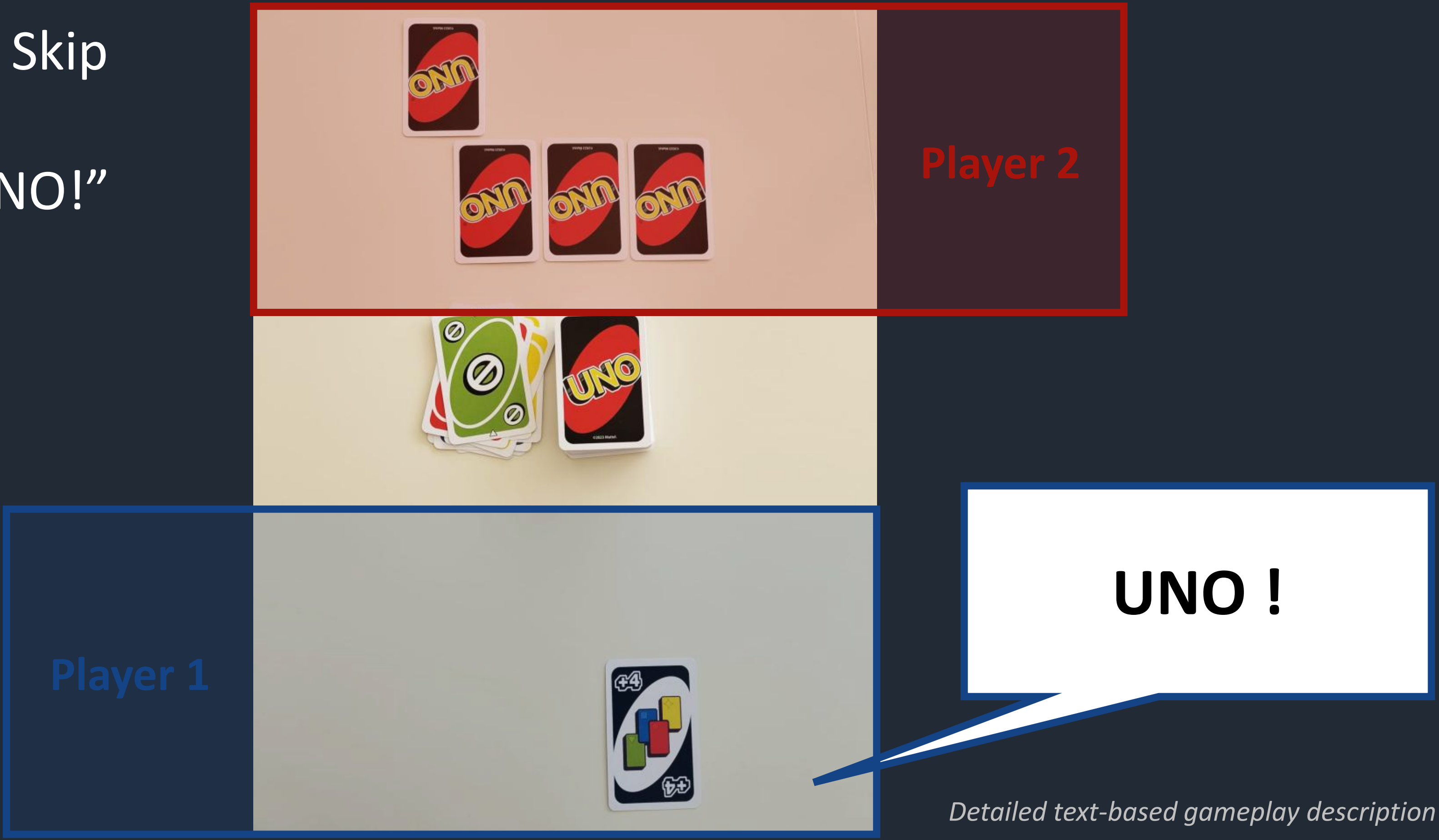
UNO® Gameplay Illustrations

Player 1 plays Green Skip

Player 1 calls out “UNO!”



Detailed text-based gameplay description in appendix

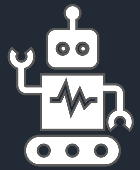
UNO® Gameplay Illustrations

Player 1 plays Wild Draw 4

Player 1 wins the match!

Detailed text-based gameplay description in appendix

Existing Research on Uno



<u>Winning Uno With Reinforcement Learning</u>

(Stanford University: O. Brown, D. Jasson, A. Swarnakar)

- ø DQN and DeepSARSA outperform random UNO agents
- ø DQN achieves highest win rate but converges slower than DeepSARSA

<u>Reinforcement learning compared to rule-based play in UNO</u>

(Stockholm University: Sara Nordström, Nils Ottosson)

- ø RL models (DQN, NFSP, DMC) outperform simple rule-based agents
- ø Rule-based strategy relies on simple heuristics and lacks adaptability

<u>The complexity of UNO</u>

(E. Demaine, M. Demaine, R. Uehara, T. Uno, Y. Uno)

- ø Formal mathematical analysis of UNO's computational complexity
- ø Proves UNO decision-making is NP-complete (single player) / PSPACE-complete (2-player competitive) in several settings

Strategies

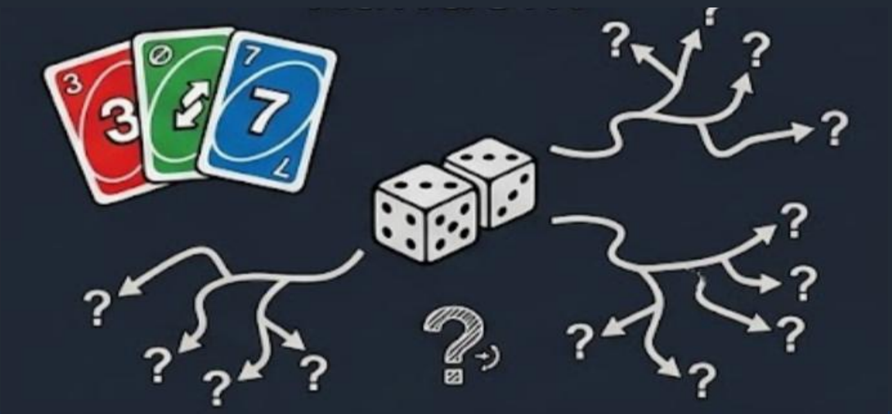
Common Strategy

- Play number cards early
- Use matching numbers to switch colors efficiently
- Save Action & Wild cards for the endgame
- Use Draw cards against low-card opponents
- Chain Skip/Reverse for surprise finishes
- Maintain colors the opponent draws on
- Force colors the opponent likely lacks
- Use draw cards when opponent near UNO
- *Control tempo by denying opponent turns*

Game-Theory Optimal (GTO)

- No single optimal strategy
- Best play depends on game state
- Decisions adapt to hidden information
- *GTO strategy is approximated using Information Set Monte Carlo Tree Search (ISMCTS)*
- ISMCTS-Handles hidden information by re-sampling the opponent's hand each simulation over a shared search tree
- Comparison benchmark

Agents



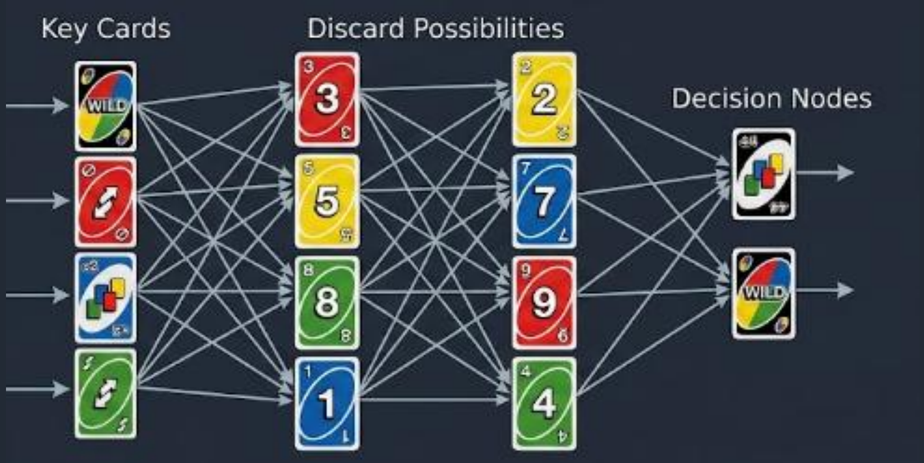
Random Agent

Greedy Rule Based Agent

VS

ISMCTS

RL Agent



*All visuals and architectures shown here are for illustrative and conceptual purposes only.

Heuristic-Based Agents

Heuristic Agents

For validation

x

Random Agent

- The agent selects a playable card completely at random from all valid options.
- If the selected card is a wild card, the agent randomly chooses a color from **Red**, **Green**, **Blue**, or **Yellow**.
- Any agent ≤ random =no learning signal.
- Random vs Random converges to ~50/50.

Greedy Rule Agent

- **Threat Mode:**

Switches when opponent has ≤2cards

Priority: +4 > +2 > Skip > Reverse > Wild > Number

- **Normal Mode:**

Save wilds; play action cards first.

Priority: Skip > Reverse> High Number

- **Wild color:**

Chooses the most frequent playable color.

Anti-Greedy Agent

- **Inverted priority**: Number >Reverse > Skip> +2 > Wild >+4
- **Wastes wilds** early with random colors and no threat awareness.
- **Greedy Agent consistently wins**, validating the heuristic strategy.
- Anti-Greedy performs near Random Agent, giving no advantage over chance.

Reinforcement Learning (RL) Agent

RL Agent – Framework

A 2-Player UNO is a sequential decision-making problem:

State	Action	Reward	Goal
Observe game state: top card, active color, hand sizes, valid plays	Decide which available card to play according to per-card scoring	Receive reward from action	Accumulate the highest total reward across the whole game

✓ Instead of been told how to play, the RL agent learns by trial and from the outcomes of the games.

RL Agent – State & Action

DQN: maximize Q-value – $Q(s,a) \leftarrow r + \gamma \max_{a'} Q(s',a')$

$State$ (s)

54-float state vector,
Top card, current color, hand sizes, etc.

$Input\ (75) \rightarrow Hidden\ layer\ (128) \rightarrow Hidden\ layer\ (64) \rightarrow$

$Action$ (a)

21-float card vector,
Candidate card's color, type, value

$Q - Value$

- Expected future reward of action a in state s
- $\rightarrow Output\ (1)$

The agent scores every legal card independently (per-card scoring) and plays the one with the highest Q-value

RL Agent – Reward and Training

Reward Design

Intermediate Reward

Placing draw Two: + 0.2 / 0.5 (Threat)
Placing draw Four: + 0.3 / 0.4 (Threat)
Hand reduced: + 0.2 * cards shed

Terminal Reward

Win =+ 1
Loss =- 1
Draw =-0.5

Training Curriculum

★ Warm-Start (Behavioral Cloning)

- Start with 5,000 Greedy vs Greedy games → pre-trains network to imitate Greedy agent
- Epsilon starts at 0.3 (instead of 1.0)

★ Prioritized Replay Buffer

- Samples by TD error — surprising turns seen more often
- Epsilon: 0.3 → 0.05 over 100,000

Phase 1
0 – 4,999
vs RandomAgent

Phase 2
5,000 – 19,999
vs Anti-Greedy Agent

Phase 3
20,000 – 100,000
vs 50% Greedy Agent & 50% Self

Information Set Monte Carlo Tree Search Agent (ISMCTS)

ISMCTS — Algorithm & UCB1

Hidden info → re-sample one possible world each iteration; share one tree across all samples.

Sample	Select	Expand	Rollout	Backprop
draw a plausible opponent hand	walk tree by UCB1 on legal children	add one child for an untried legal action	play until terminal or depth	push result up; visits & availability

What does ISMCTS mean?

Single-Observer: model only player 0's information set;
Information Set: the set of true game states consistent with what we've seen.
Monte Carlo TS :tree search guided by simulated playouts (the 5-step loop above).
vs determinized UCT :fix one world before search; entire tree is biased toward that sample. ISMCTS resamples per iteration, averaging over uncertainty.

UCB1 — availability-normalized

ucb1(c) = wins/visits
+ c · √(log(availability) / visits)

availability = times node was legal across determinizations
visits = times node was actually selected
availability ≥ visits (always)

vs UCT: uses log(parent.visits) — ignores legality across determinizations

UCB (Upper Confidence Bound) A formula that balances trying new options vs sticking with what's worked.
UCT (Upper Confidence bounds applied to Trees) Applying UCB to tree search. Introduced by Kocsis & Szepesvári in 2006.

ISMCTS — Experimental Design

Vary one parameter at a time; isolate each effect against the same rule-based opponent(500 games).

c — exploration constant

{ 0.35, 0.7, 1.0, 1.41, 2.0, 3.0 }

Centred on $\sqrt{2} \approx 1.41$ (UCT theoretical optimum); halves and doubles around it.

n — num_simulations — search budget

{ 25, 50, 100, 200, 500 }

Doubling schedule to find the gain-curve elbow

d — rollout_depth — playout horizon

{ 10, 20, 30, 50, 100, 150 }

Around UNO's ~31 plies / player; heuristic fallback past the depth limit.

π — rollout_policy — playout π

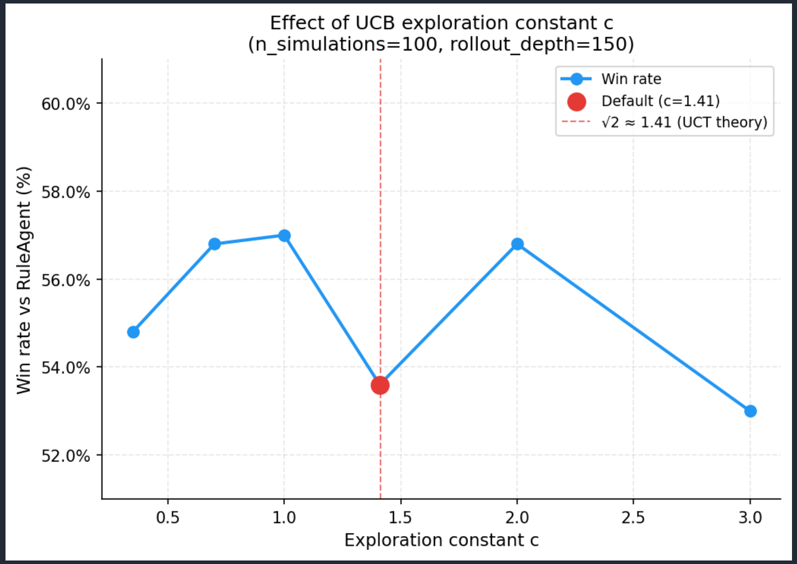
{ random, rule }

Random gives unbiased trajectories; rule is a lightweight priority heuristic.

heuristic fallbackscore = 0.5 + 0.2 × (opponent_hand - my_hand) / (opponent_hand + my_hand)

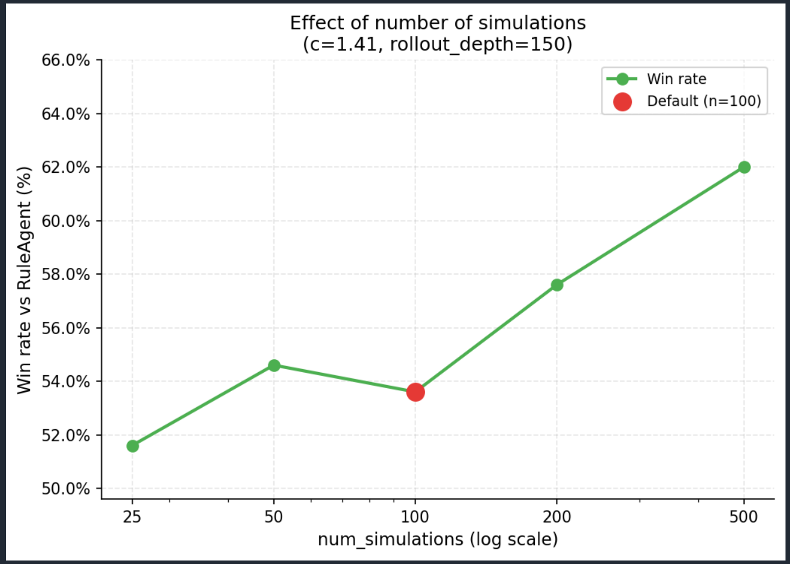
ISMCTS — 1D Sweep Results

c



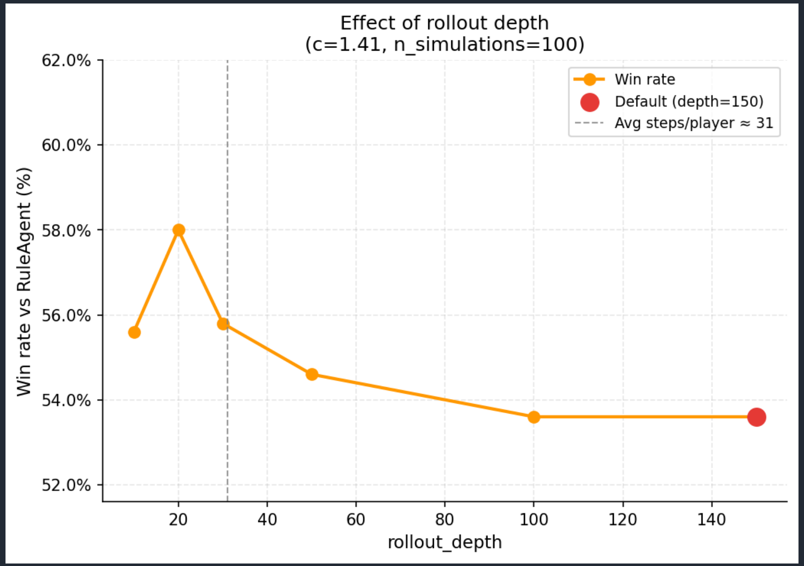
Peak 57.0 % at c = 1.0
flat in [0.7, 2.0]

num_simulations



Monotonic rise 51.6 % → 62.0 %
most sensitive

rollout_depth



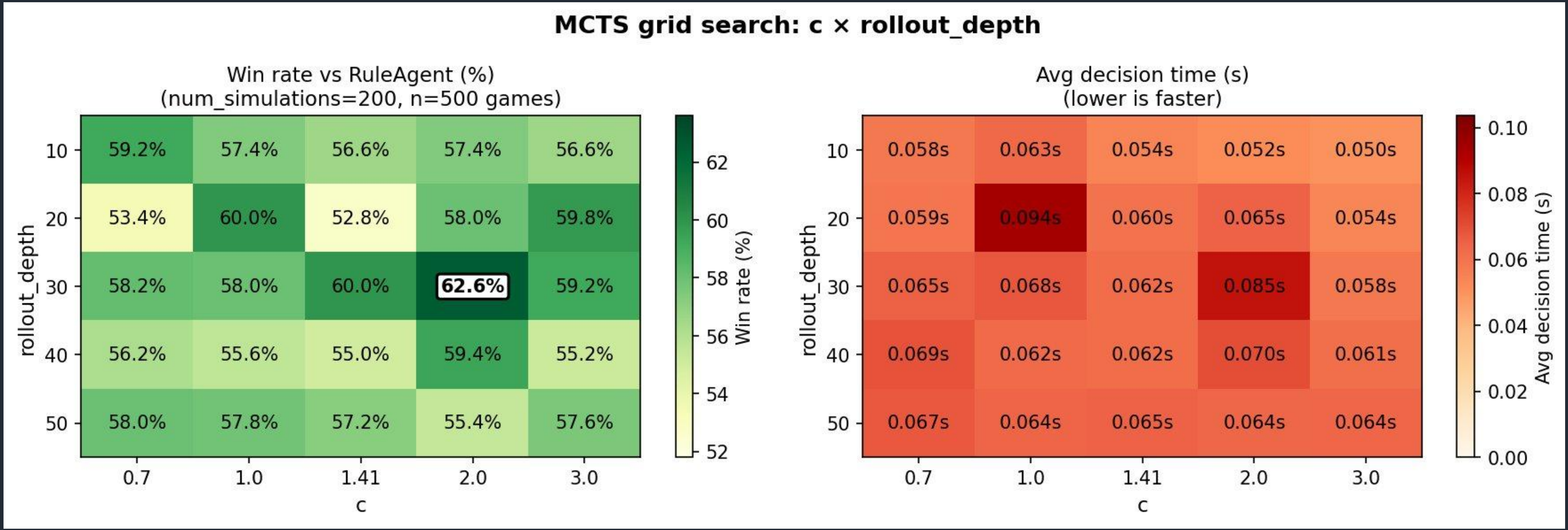
Peak 58.0 % at depth = 20
degrades after

Take-away

1D best: **c = 1.0, num_simulations = 200, depth = 20**→ 58.8 % vs Rule, 64.1 % vs Random (1000 games).

ISMCTS — Grid Search

One-at-a-time sweeps can't see how parameters interact; we run a 2-D grid to find out.



Take-away

2D best: **c = 2.0, num_simulations = 200, depth = 30** → 59.8 % vs Rule, 63.3 % vs Random (1000 games).

Analysis & Takeaways

Analysis Logic

We evaluate agents across three dimensions :

1 - Performance
Win Rate

2 - Efficiency
Surviving Rounds

3 - Fairness
First-Mover Advantage

Proof of reliability
Evaluation Framework

ISMCTS as the Near-GTO Baseline

- All agents compared against ISMCTS
- ISMCTS simulates thousands of future game states per turn, serving as the closest approximation to a GTO strategy

Statistical Significance Testing
p-value & confidence interval

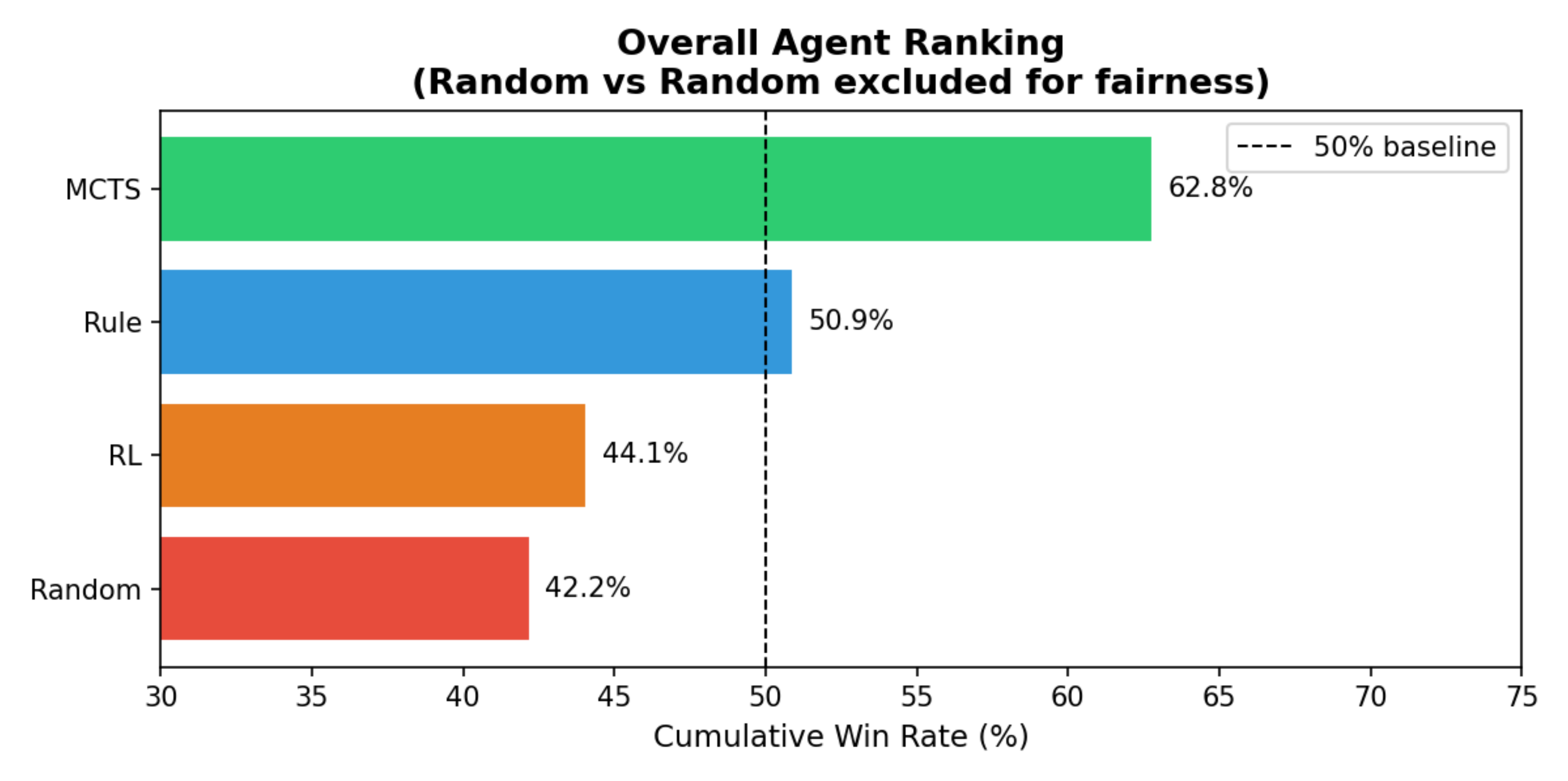
- Win/loss → Binomial test;
- Surviving rounds → Mann-Whitney U test

10,000 games per matchup

- 95% confidence interval narrows to ±1%
- Reliably detect small but real differences in agent performance

Overall Agent Ranking

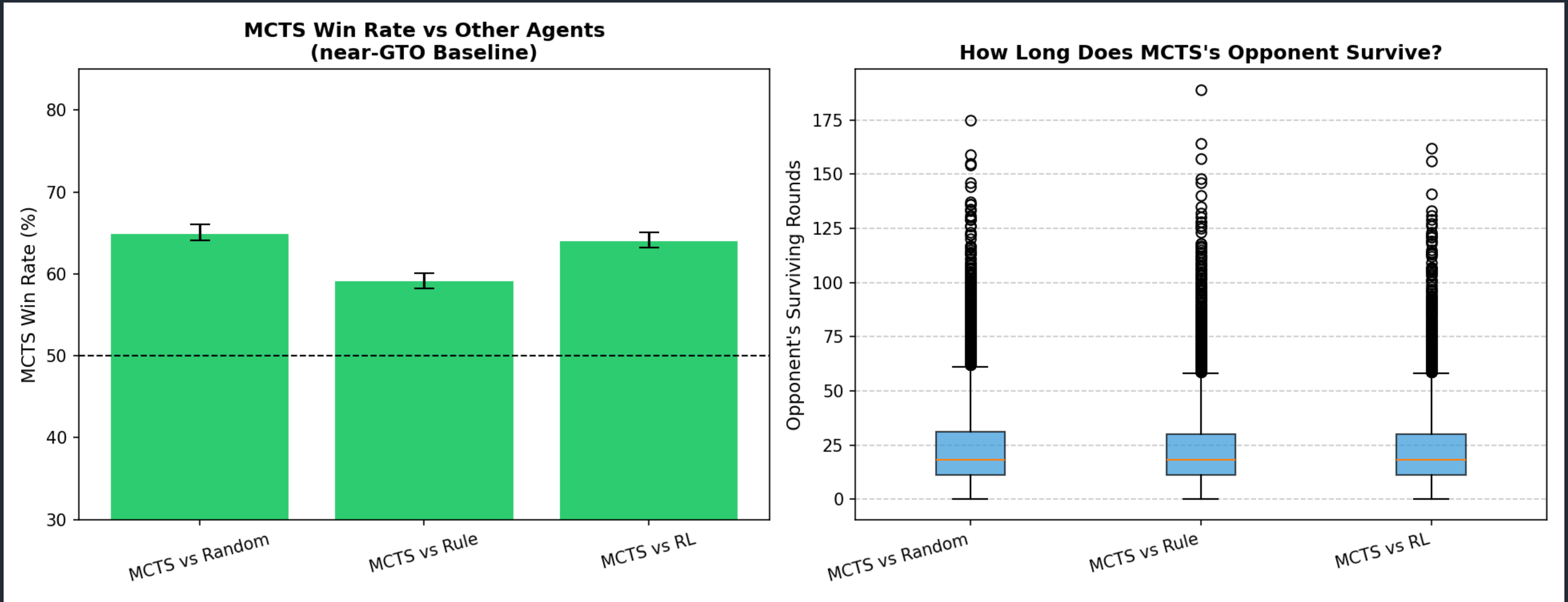
- **ISMCTS significantly outperforming all others.** No evidence of cyclic dominance is observed; instead, the agents exhibit **a stable linear ranking: ISMCTS > Rule > RL > Random.**
- ISMCTS achieves the highest win rate (~62.8%), followed by Rule (~50.9%), while RL (~44.1%) and Random (~42.2%) perform below the baseline.



Ag	TotalWins	TotalGames
ISMCTS	18,834	30,000
Rule	15,256	30,000
RL	13,240	30,000
Random	12,670	30,000

ISMCTS VS Other Agents

- **ISMCTS** consistently outperforms all agents.
- It shows a clear advantage over Random and RL (~65%), while the margin against the rule-based agent is smaller (~58–60%), indicating that **rule-based strategies partially approximate optimal play.**
- Opponent survival rounds are similar across agents, with only minor differences, suggesting that **performance differences are primarily reflected in win rates rather than game duration.**

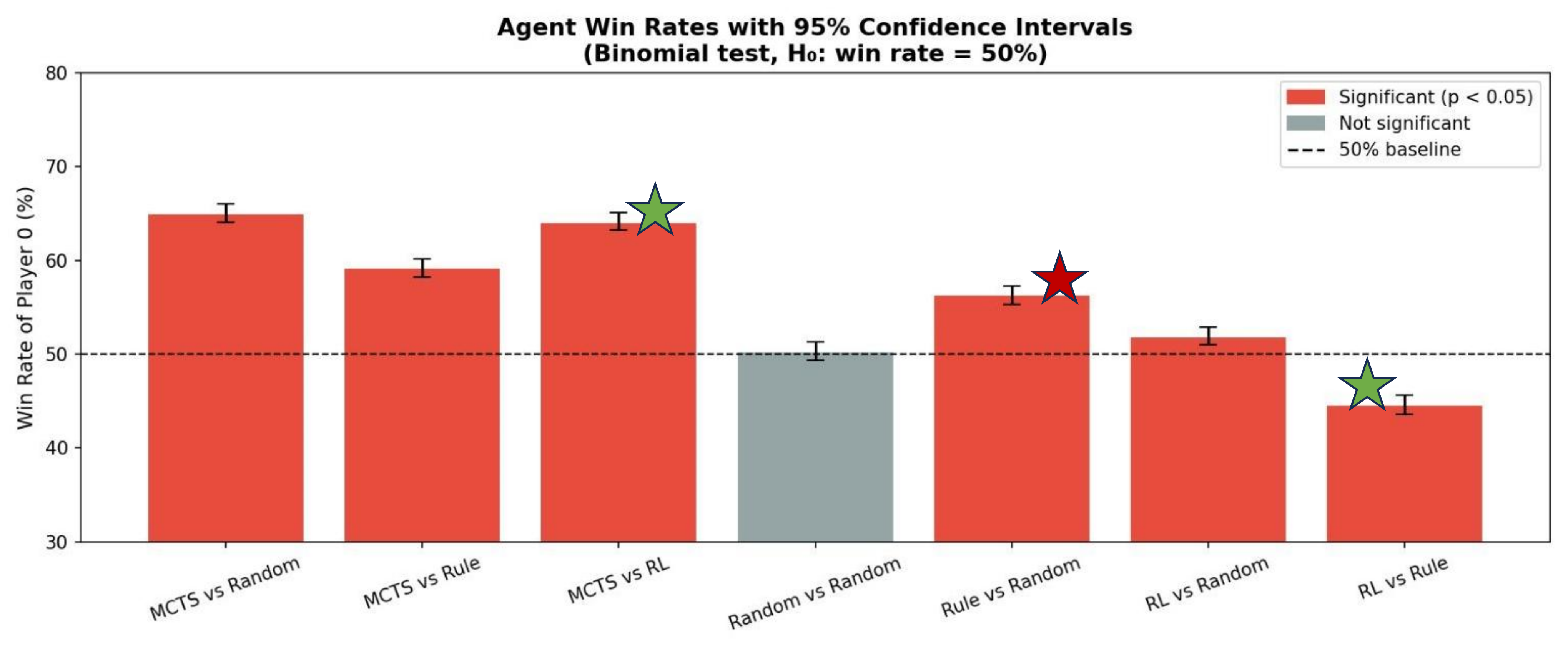


1 - Performance

Win Rate

Statistical testing confirms that ISMCTS shows statistically significant gains over all other agents.

- **RL exhibits consistent underperformance**, confirming that its **lower win rates reflect genuinely weaker policies rather than random variation.**
- **Rule outperforms Random**, indicating that **Rule-based heuristics provide a reliable improvement over uninformed play.**



2 - Efficiency

Surviving Rounds

ISMCTS consistently achieves shorter opponent survival times, indicating faster, more decisive gameplay with statistically reliable advantages.

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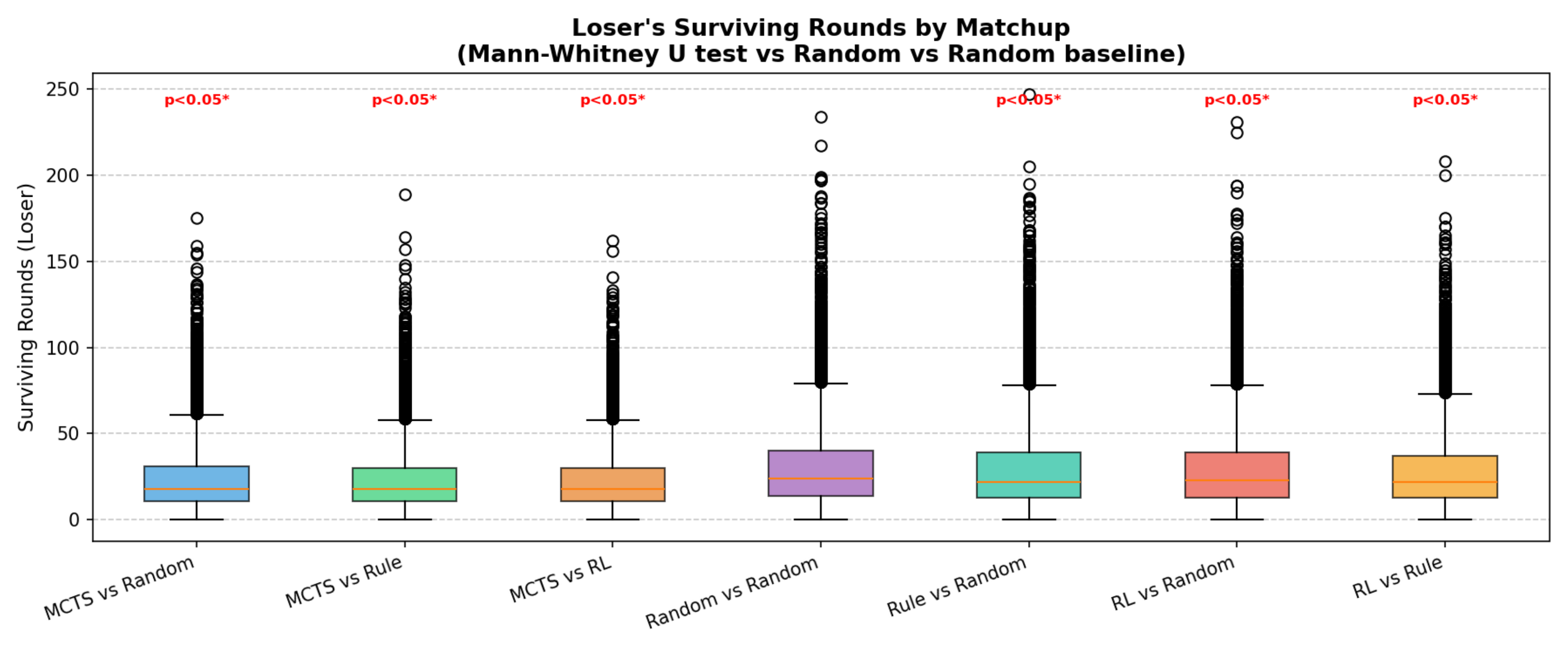
- **Rule-based and RL agents** show significantly longer opponent survival times (~29 rounds vs ~23 for ISMCTS), suggesting **weaker control over game progression**.
- All pairwise differences are statistically significant ($p < 0.05$), confirming that **survival rounds reliably reflect performance differences rather than random variation.**

Matchup	Surv_Avg	Surv_Std
ISMCTS vs Random	23.7	18.3
ISMCTS vs Rule	23.1	17.6
ISMCTS vs RL	23.0	17.0
Rule vs Random	29.4	24.2
RL vs Random	29.5	23.1
RL vs Rule	28.6	22.4



3 - Fairness

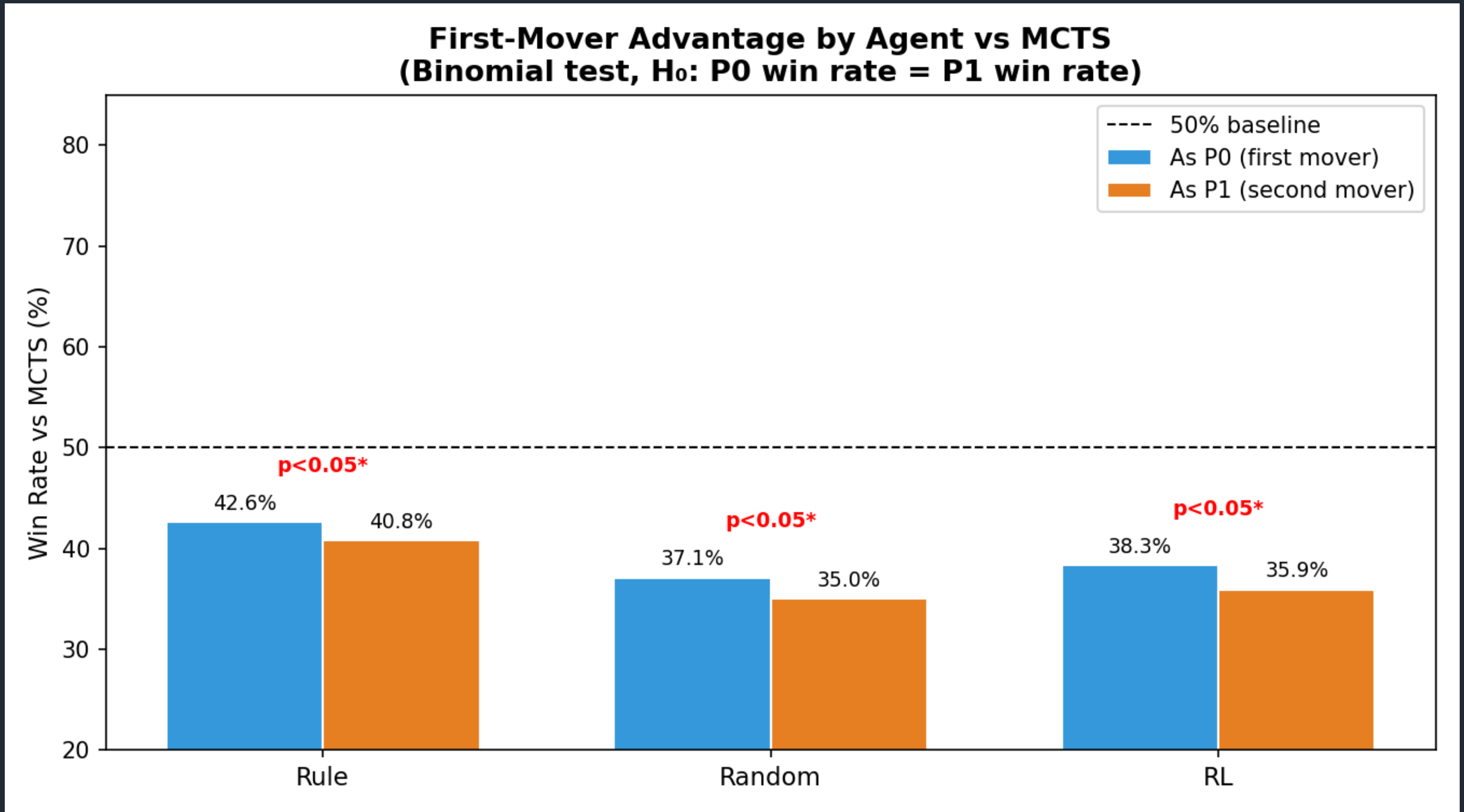
First-Mover Advantage

All agents exhibit a statistically significant first-mover advantage, indicating a systematic bias favoring the starting player.

x UNO

- The magnitude of this advantage increases from Rule (1.2) to Random (2.1) and RL (2.4), suggesting that **weaker or less structured agents are more sensitive to turn order.**
- This implies that **stronger strategies are more robust to positional bias, while weaker agents rely more heavily on first-move advantage to perform competitively.**

Agent	Advantage= P0 win rate – P1 win rate
Rule	1.2
Random	2.1
RL	2.4



Takeaways

Based on the results across performance, efficiency, and fairness, we can summarize the key findings as follows:

1. ISMCTS > Rule > RL > Random

ISMCTS is the strongest agent, consistently outperforming others with statistically significant advantages over Random and RL, and Rule.

2. Performance differences are driven primarily by win rates rather than game duration, with ISMCTS achieving faster and more decisive outcomes while maintaining similar survival variability.

3. Game dynamics are not perfectly fair: a statistically significant first-move advantage exists for all agents, with weaker strategies (Random, RL) being more sensitive to turn order than stronger ones.


UNO
00:11:31
PLAYER 1

Compete against our UNO agents!

Scan the QR-code!

Or visit the link : https://uno-matchup.streamlit.app/

Limitations

- The current system only supports 2-player UNO, limiting multiplayer strategy complexity.
- The RL agent uses a relatively simple DQN architecture, which restricts long-term strategic learning.
- The ISMCTS agent achieves the best performance but requires significantly higher computation time.
- Hidden information is approximated through sampling, so opponent behavior cannot be modeled perfectly.
- Evaluation mainly focuses on win rates and average turns rather than broader strategic metrics.

Future Scope

- Extend the environment to support full multiplayer UNO gameplay.
- Improve the RL agent using advanced methods such as PPO, Double DQN, or Transformer-based models.
- Combine ISMCTS with neural networks to create stronger hybrid agents.
- Add opponent modeling so agents can adapt to different play styles.
- Expand benchmarking with Elo ratings, statistical testing, and richer evaluation metrics.

References

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Thank you for your attention!

Now it's *your* turn

ø Do you have any questions?

ø Feedback?

Appendix

UNO® Gameplay

- On your turn, <u>you try to get rid of all your cards </u>by playing ONE CARD onto the Discard Pile.
- If you HAVE a matching card, you may PLAY IT on the Discard Pile.
  - You can only play a card if it matches at least one attribute of the top card on the Discard Pile: it’s color, number, or symbol
  - If you played an action card it does something special (see next slide)
- If you DO NOT HAVE a matching card, DRAW ONE CARD from the draw pile
  - If your *new* card can be played (see above), then you may play it now.
  - You may choose to draw a card, even if you have a playable card in your hand.
- Once you have only one card in your hand you must yell “UNO”.
  - If someone catches you and calls out “UNO” before you, then you must draw two cards.

Another UNO® Match

Wild Draw Four – Challenge Mechanic

Wild Draw Four

This card matches anything so you can play it no matter what card is on the Discard Pile. However, there's a catch: you can only play a Wild Draw 4 if you <u>DO NOT HAVE ANY CARD IN YOUR HAND THAT MATCHES THE COLOR OF THE DISCARD PILE</u>.

When played, the next player has two options: Draw 4 cards and lose their turn OR challenge.

If the player challenges you, then you must show them your entire hand to confirm whether or not you have a card that matches the color of the discard pile – Wild cards are considered a match, too.

- If you <u>DO NOT</u> have a card that matches color: the challenger draws 6 cards instead of 4 and loses their turn.
- If you <u>DO</u> have a card that matches color: YOU must draw 4 cards and they draw none.

This card is also a Wild card, so you get to choose the color that continues play (regardless of the outcome of any challenge).

Wild Draw Four – Illegal Plays

- An illegal play occurs when you have a card matching the color of the card on the discard pile OR you have a wild card.
- As a result, your Wild Draw Four (+4) card could be successfully contested by your opponent.

Illegal play because the player is holding a color matching card (yellow 3) in their hand.

Illegal play because the player is holding a wild card in their hand.

A wild card is considered a match, too.

Wild Draw Four – Legal Plays

- A legal play of the Wild Draw Four (+4) card is any play when the player does not hold a card matching the color of the top card on the discard pile.
- Challenging these plays would result in +6 draw for the opponent.

Even though the player *could* play the blue 9 this play is legal.

This is because the rule only regards color, not number or symbol.